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North America								
Canada 1220	82,550	77,833	257,845,559	233,282,941	8,937	8,039	28,181,268	22,965,296
Mexico 2010	203,709	250,547	496,188,629	628,915,910	22,295	26,587	54,193,116	64,754,079
Total	286,259	328,380	754,034,188	862,198,851	31,232	34,626	82,374,384	87,719,375
Caribbean								
Bermuda 2320	871	665	5,402,663	4,891,077	65	56	442,283	384,429
Bahamas; The 2360	1,274	3,307	3,500,629	8,461,241	290	261	561,031	539,101
Jamaica 2410	2,013	3,123	2,119,514	4,446,927	116	391	155,831	691,771
Turks and Caicos Islands 2	43	44	456,894	346,982	1	8	11,256	26,094
Cayman Islands 2440	1,719	225	5,203,439	694,571	176	0	432,572	0
Haiti 2450	18	83	74,086	129,959	0	0	0	0
Dominican Republic 2470	2,058	963	3,490,732	2,143,918	107	129	211,065	381,574
Leeward-Windward Islands	424	466	1,267,391	1,243,525	19	71	75,109	169,179
Barbados 2720	533	169	2,070,054	976,904	16	27	39,148	221,399
Trinidad and Tobago 2740	306	126	924,478	279,263	44	28	87,052	40,071
Netherlands Antilles 2770	448	404	1,533,095	1,337,232	32	34	127,634	89,449
French West Indies 2830	94	14	399,621	30,795	9	12	68,959	27,116
Total	9,802	9,589	26,442,596	24,982,394	875	1,019	2,211,940	2,570,183
Central America								
Guatemala 2050	1,244	1,028	2,965,718	3,258,465	135	87	280,076	277,937
Belize 2080	36	4	65,121	5,516	0	0	0	0
El Salvador 2110	79	159	161,299	419,593	17	27	7,315	94,386
Honduras 2150	554	203	912,046	437,944	26	0	64,410	0
Nicaragua 2190	78	27	148,794	89,039	7	4	32,888	16,490
Costa Rica 2230	559	1,119	883,540	1,738,845	54	141	65,568	216,368
Panama 2250	2,118	775	2,800,392	1,940,522	107	88	162,713	219,058
Total	4,667	3,314	7,936,910	7,889,924	348	348	612,970	824,239
South America								
Colombia 3010	361	631	472,174	594,921	31	47	43,523	28,620
Venezuela 3070	2,493	1,027	2,829,830	2,365,392	196	137	197,269	425,345
Guyana 3120	0	16	0	16,677	0	11	0	11,052
Suriname 3150	0	5	0	14,614	0	0	0	0
Ecuador 3310	866	57	927,798	100,090	0	1	0	6,647
Peru 3330	1,662	863	1,729,705	1,033,412	48	297	42,000	266,542
Bolivia 3350	1	0	4,316	0	0	0	0	0
Chile 3370	5	195	38,972	279,403	0	46	0	47,713
Brazil 3510	3,636	240	3,759,483	420,791	19	50	60,416	44,001
Uruguay 3550	213	427	531,422	917,826	1	24	11,131	35,758
Argentina 3570	3,000	2,257	3,733,462	3,647,807	117	300	199,604	336,480

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Total	12,236	5,718	14,027,162	9,390,933	411	914	553,943	1,202,158
European Union								
Sweden 4010	1,246	11	1,284,566	123,749	0	0	0	0
Finland 4050	554	13	564,796	140,189	47	0	34,000	0
Denmark 4090	234	131	1,045,165	923,224	4	0	22,515	0
United Kingdom 4120	1,074	831	2,767,638	372,604	13	5	31,470	6,500
Ireland 4190	14	20	43,553	8,880	0	0	0	0
Netherlands 4210	1,736	361	4,626,328	1,726,345	100	2	425,653	19,380
Belgium-Luxembourg 4230	9,783	4,456	4,215,347	3,350,193	1,719	22	494,148	190,920
France 4270	990	555	1,248,396	471,841	39	14	48,600	72,625
Germany 4280	8,058	6,189	6,103,699	4,736,152	1,343	285	684,137	424,221
Austria 4330	0	300	0	107,660	0	8	0	23,234
Spain 4700	331	143	1,588,609	726,885	34	7	187,630	51,928
Portugal 4710	35	39	126,290	108,961	0	9	0	79,316
Italy 4750	90	183	419,770	1,119,596	6	4	17,307	48,765
Greece 4840	407	154	987,885	547,850	88	2	152,206	15,609
Total	24,550	13,386	25,022,042	14,464,129	3,392	358	2,097,666	932,498
Other West Europe								
Iceland 4000	0	1	0	6,214	0	1	0	6,214
Norway 4030	0	8	0	7,772	0	0	0	0
Switzerland 4410	667	803	6,007,931	5,189,985	29	74	370,191	406,695
Malta & Gozo 4730	135	38	1,333,850	450,911	0	0	0	0
Total	802	850	7,341,781	5,654,882	29	75	370,191	412,909
Former Soviet Union								
Estonia 4470	1,721	0	985,024	0	201	0	105,137	0
Latvia 4490	14,539	4,942	8,340,820	3,016,783	1,551	0	835,088	0
Lithuania 4510	0	24	0	15,000	0	0	0	0
Russian Federation 4621	21,158	42,800	23,144,595	35,390,761	3,063	4,510	2,656,266	4,314,180
Ukraine 4623	15	0	19,987	0	0	0	0	0
Armenia; Republic of 4631	2	0	8,000	0	0	0	0	0
Georgia; Republic of 4633	8	158	3,600	207,417	0	0	0	0
Total	37,442	47,923	32,502,026	38,629,961	4,815	4,510	3,596,491	4,314,180
Eastern Europe								
Hungary 4370	14	40	161,680	198,382	0	0	0	0
Poland 4550	5,162	4,461	3,927,117	2,685,854	660	701	519,222	424,572
Slovenia 4792	5	0	37,734	0	0	0	0	0
Macedonia (Skopje) 4794	23	43	22,300	34,858	0	0	0	0
Albania 4810	48	0	56,405	0	0	0	0	0
Romania 4850	266	875	261,772	734,155	88	104	100,652	102,947

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Bulgaria 4870	229	179	246,782	180,131	24	0	24,271	0
Total	5,746	5,598	4,713,790	3,833,380	772	805	644,145	527,519
Middle East								
Turkey 4890	47	3	120,464	25,334	0	0	0	0
Lebanon 5040	192	69	460,154	291,279	70	3	160,751	6,976
Israel 5080	781	837	1,041,409	851,810	72	190	53,685	159,949
Jordan 5110	91	21	129,858	209,118	0	0	0	0
Kuwait 5130	290	265	1,284,737	2,202,837	16	48	124,339	453,078
Saudi Arabia 5170	1,131	721	3,182,662	3,737,463	107	59	248,811	479,046
Qatar 5180	41	34	332,926	337,308	25	9	197,704	115,778
United Arab Emirates 5200	585	490	2,460,588	3,742,875	100	138	434,272	680,673
Yemen 5210	0	6	0	61,862	0	5	0	46,191
Oman 5230	3	14	20,484	78,486	0	9	0	46,310
Bahrain 5250	73	504	407,392	2,325,943	1	162	4,948	709,511
Total	3,235	2,963	9,440,674	13,864,315	391	623	1,224,510	2,697,512
North Africa								
Morocco 7140	0	18	0	68,122	0	0	0	0
Egypt 7290	26,975	27,780	24,080,443	23,361,526	3,425	3,137	3,348,152	3,000,804
Total	26,975	27,798	24,080,443	23,429,648	3,425	3,137	3,348,152	3,000,804
Other Africa								
Equatorial Guinea 7380	0	2	0	8,634	0	0	0	0
Senegal 7440	0	104	0	134,725	0	0	0	0
Guinea 7460	1	3	11,999	18,910	0	0	0	0
Cote d"Ivoire 7480	1,599	2,205	1,392,493	1,093,276	141	192	65,206	96,751
Ghana 7490	176	8	245,114	31,394	0	4	0	16,320
Nigeria 7530	8	14	26,068	29,449	0	0	0	0
Gabon 7550	389	1,241	323,486	777,965	25	216	66,609	165,000
Angola 7620	860	836	777,832	746,509	82	0	81,766	0
Congo (Brazzaville) 7630	124	0	101,846	0	25	0	15,777	0
Zaire 7660	0	30	0	25,612	0	0	0	0
Kenya 7790	66	1	66,475	14,762	41	0	34,575	0
South Africa; Republic of 79	551	284	423,670	1,829,692	100	0	95,881	0
Namibia 7920	0	30	0	64,623	0	2	0	3,981
Total	3,775	4,759	3,368,983	4,775,551	415	414	359,814	282,052
South Asia								
India 5330	23	0	13,460	0	0	0	0	0
Bangladesh 5380	0	0	2,512	2,840	0	0	2,512	0
Total	24	0	15,972	2,840	0	0	2,512	0
Other Asia								

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Thailand 5490	150	236	732,661	1,672,850	25	44	137,674	263,864
Vietnam 5520	5	48	61,615	233,019	0	0	0	0
Laos 5530	0	18	0	57,716	0	0	0	0
Cambodia 5550	22	4	29,065	15,841	18	0	16,400	0
Malaysia 5570	998	345	1,938,273	2,121,638	0	61	0	345,754
Singapore 5590	1,534	715	5,454,475	4,473,930	97	72	554,263	521,510
Indonesia 5600	6,133	6,278	6,386,957	7,370,158	357	2,229	715,076	2,044,894
Brunei 5610	47	0	34,000	0	0	0	0	0
Philippines 5650	550	788	3,403,919	2,920,221	132	104	1,005,630	723,184
China; Peoples Republic of	8,085	7,704	12,872,153	18,734,033	633	1,206	1,471,466	2,753,284
Korea; Republic of 5800	136,078	110,052	440,573,277	282,293,793	8,978	17,611	28,796,954	44,624,264
Hong Kong 5820	22,518	18,155	64,814,352	50,435,134	2,383	2,370	7,771,312	8,089,285
Taiwan 5830	17,766	11,226	54,060,745	39,472,182	1,210	1,287	4,476,651	4,921,381
Japan 5880	461,022	458,537	1,541,265,014	1,439,760,594	41,075	44,925	127,518,707	142,500,817
Total	654,907	614,106	2,131,626,506	1,849,561,109	54,908	69,908	172,464,133	206,788,237
Oceania								
Australia 6020	2,889	622	4,143,400	1,204,285	28	13	69,957	64,340
New Zealand 6140	72	16	158,917	57,411	0	0	0	0
Western Samoa 6150	7	5	27,802	13,516	0	0	0	0
French Pacific Islands 6410	167	25	1,033,384	172,496	0	2	0	26,394
Marshal Islands 6810	405	153	705,455	467,066	18	21	40,194	68,791
Micronesia; Federated States	180	92	301,838	321,215	2	4	4,986	23,615
Palau 6830	266	172	552,244	630,934	23	14	87,211	65,992
Other Pacific Islands; NEC 6	40	1	58,710	3,201	15	0	22,506	0
Total	4,025	1,086	6,981,750	2,870,124	85	54	224,854	249,132
Grand Total	1,074,446	1,065,470	3,047,534,823	2,861,548,041	101,096	116,790	270,085,705	311,520,798

COMMODITY AGGREGATES/GROUPS AND MEMBER CODES
Beef, F/CH/FZ/Prep/Pres/Var Meat, Export

X0201100010	CRC,1/2CRCVL F/C
X0201100090	CARCASSES/HALF-C
X0201203000	BF PRSDW/B FR/CH
X0201203550	BF PRSDW/B FR/CH
X0201206000	BFW/B NTPRSD F/C
X0201303000	BF PRSD WO/B,F/C
X0201303550	BF PRSD WO/B,F/C
X0201306000	BFWO/B NPRSD F/C
X0202100010	CRC,1/2CRC VL FZ

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X0202100090	CARCASSES/HALF-C								
X0202203000	BF PRSD W/B F/CH								
X0202203550	BF PRSD W/B F/CH								
X0202206000	BFW/B CRC,FRZ								
X0202303000	BF PRSD WO/B FRZ								
X0202303550	BF PRSD WO/B FRZ								
X0202306000	BFWO/B, CRC,FRZ								
X0210200000	BF,SLT,BR,DRD,SM								
X1602509020	PREP MEALS,BEEF								
X1602509500	BEEF,PREP,PRES								
X0206100000	BOV OFL,ED,FR,CH								
X0206210000	BOV TNGS,ED FRZ								
X0206220000	BOV LVRS,ED FRZE								
X0206290010	BOV OFL,FZ,HEART								
X0206290020	BOV OFL,FZ,KDNEY								
X0206290030	BOV OFL,FZ,BRAIN								
X0206290040	BOV OFL,FZ,SWTBD								
X0206290050	BOV OFL,FZ,LIPS								
X0206290090	BOV OFL,FZ,OTHER								
X0504000050	BEEF TRIPE, FZ								
X0504000070	BF INT,FZ,X SSGC								

9/02/2008 This Week in Congress

This Week in Congress

September 02, 2008

Dear Friend,

Welcome to "This Week in Congress." I hope you find this newsletter useful.

Talking to Senior Citizens about Improvements to Medicare

On Friday, I visited the Senior Center in Hays to talk to senior citizens about passage of the Medicare Improvements for Patients and Providers Act, H.R. 6331. This legislation, which passed the House in June and has since become law, eliminated scheduled reductions in Medicare physician payments, extended physical therapy caps and delayed reductions in pharmacy reimbursement rates. Passage of this legislation was critical for Kansans as it allows hospitals, doctors, physical therapists and pharmacists to continue serving the more than 114,000 Medicare recipients in the First District.

I was joined at the Senior Center by David Wilson, president of AARP Kansas; Dr. Michael Machen of Quinter, president of the Kansas Society of Health System Pharmacists; Jim Rorstrom of Hays; Fred Lucky, with the Kansas Hospital Association; and Dave Sanderson of Salina, with the Kansas Physical Therapy Association. These experts were able to explain to folks how the legislation affected their profession and answered questions about what the new law means for Medicare recipients. Also discussed during the event was the AARP "Divided We Fail Pledge." This is a congressional pledge asking the nation's leaders to commit to working in a bipartisan way to provide Americans with actions and answers on health and long-term financial security. This is a pledge I strongly support and have signed my name to.

I enjoyed speaking to many of the seniors gathered for lunch, as well as Ernie Kutzley of Topeka, AARP Kansas; Mary Tritsch of Topeka, AARP Kansas; Gerald and Dianna Schmitt of Wichita, AARP volunteers; Alberta Klaus of Hays, president of AARP chapter in Hays; and Dan Morin with the Kansas Medical Society. Special thanks to Alice Herman, director of the Meal Site, for hosting the forum. Click here to view photos from the event.

Visiting North Central Kansas

On Monday, I visited several communities in north central Kansas. While in Blue Rapids, I visited the State Bank, Yungeberg Drug, the Post Office, Napa Auto Parts and several city offices. After my visit in Blue Rapids, I traveled to Marysville to attend the Rotary Club meeting. Thanks to Marysville Rotary president, Kate Manley.

Also while I was in Marysville, I made a trip into the Marysville Advocate, a 122 year-old community newspaper. There, I saw Sarah Kessinger, the new editor for the Marysville Advocate. She recently took over the paper from her parents, Howard and Sharon. I commend her and the Kessinger family for working hard to keep this local publication serving their community.

I also visited Hanover to talk with community members. There I visited with employees at Citizens State Bank, Hanover Professional Pharmacy, the Post Office, Hanover Hospital, Andy's Hardware and Lumber, Crome's Market, Blue Valley Insurance and Rural Water District #1.

In my visits with local residents, I was again reminded of my belief that Congress must immediately take up a comprehensive energy plan to assist those families who are struggling with gas prices.

Learning about Cancer Prevention and Treatment with the American Cancer Society in Alma

On Monday, I met with Tonia Carlson of Paxico and Lisa Benlon of Overland Park to discuss ways in which Congress can help increase cancer screening and prevention treatments. Ms. Carlson is the grassroots leader of the First Congressional District for the American Cancer Society. Ms. Benlon is a former State Representative who is now with the American Cancer Society

One of the issues we talked about was colon cancer. This disease is one of the leading preventable causes of death in the United States, with a five-year survival rate of nearly 90 percent when detected early. I am a sponsor of the Colorectal Cancer Early Detection, Prevention and Treatment Act, which would help those most likely at risk of colon cancer to receive screenings and treatment at an earlier age.

Speaking to Farm Bureau Members in Wabaunsee and Osage Counties

This week, I spoke at the Wabaunsee County Farm Bureau Annual Meeting in Alma and at the Osage County Farm Bureau Annual Meeting in Carbondale. I spoke to members about the 2008 Farm Bill and the need for Congress to pass comprehensive energy legislation. Attending Farm Bureau meetings across our state allows me to learn more about issues important to farmers and ranchers.

I appreciated the opportunity to meet many officers and members. Special thanks to Wabaunsee County Farm Bureau President Ken Flagler and Osage County Farm Bureau President William Prescott for inviting me to speak. Click here for photos from the Osage County meeting.

Speaking to the Kansas Agribusiness Retailers Association in Manhattan

On Tuesday, I joined the Kansas Agribusiness Retailers Association (KARA) for breakfast at their annual conference in Manhattan. Before introducing Senator Pat Roberts, I explained my plan for solving our nation's energy challenges and also talked about the 2008 Farm Bill. Both Senator Roberts and I talked about how Congress has become more urban and explained the difficulty in getting members from urban areas to understand the importance of agriculture policy in rural America.

I also expressed my condolences on the passing of one of KARA's outstanding leaders, James B. "J.B." Pearl. Mr. Pearl was the founder of J.B. Pearl Sales and Services in St. Marys, which was recognized in 2003 as the National Ag Retailer of the Year. My thoughts and prayers are with Mr. Pearl's family and friends. Thank you to President Tom Tunnel and KARA for inviting me to speak.

Observing Chapman Rebuilding and Recovering

On Tuesday, I visited Chapman to observe the town's recovery efforts and to visit with their community members. Chapman received extensive damage when a tornado touched down on the evening of June 12, 2008. While there this week, I visited the Unified School District (U.S.D.) 473 offices, City Hall, Chapman Public Library where I spoke with the Library Director Carol Frasure, Astra Bank, Londeen Hardware & Furniture Store, Insurance Store, Chapman Senior Center and Shamrock Cafe.

I was very interested in visiting Chapman High School. I believed it was important to congratulate the school's administration and teachers on beginning class this fall. The school faced tremendous circumstances following the tornado and I wanted to commend them for starting school up as scheduled.

Visiting Garden City

Garden City Community College: On Thursday, I made a visit to Garden City Community College. I had a good conversation with Dr. Carol Ballantyne, the president of the college. On my visit, I learned about the opportunity the planned Sunflower coal-fired power plant expansion presents for training construction trades at the college. We also talked about the Higher Education Act, education of health care professionals for southwestern Kansas and the TRIO program.

St. Catherine Hospital: I also toured St. Catherine Hospital (SCH) in Garden City. SCH is an affiliate of Catholic Health Initiatives, the largest Catholic not-for-profit health organization in the country. It is licensed for 132 beds, is an accredited regional health care center and has a medical staff that consists of specialists in family practice, internal medicine, obstetrics and gynecology, gastroenterology, ophthalmology, orthopedics, pathology, pediatrics, podiatry, psychiatry, pulmonary medicine, oncology, radiology, general and thoracic surgery and urology.

During my visit, I had a great discussion with Scott Taylor, SCH President & CEO. We discussed Medicare billing and reimbursements, pricing privileges for prescription drug programs, pay for performance of providers and charity care. Following my visit with Mr. Taylor, John Yox, SCH Senior Vice President and Bonnie Peters, SCH Vice President for

Patient Services, gave me a tour of the hospital facilities where I was also able to meet a number of doctors, nurses and staff. Thanks to Beth Koksal for coordinating my visit. Click here to view photos of my visit.

Attending Fort Hays State University Foundation Board of Trustees Meeting

I serve on the Fort Hays State University (FHSU) Foundation Board of Trustees. Each year, the foundation helps make a college education a reality for young people by distributing approximately $1.4 million in scholarships. On Friday, I attended a board meeting in Hays about promoting educational excellence. I enjoyed meeting with the other board members and President and CEO Tim Chapman. Congratulations to incoming chair Dr. John Tomlinson and job well done to outgoing chair Kerry McQueen.

Annual Kansas State Fair Beginning This Week

The annual Kansas State Fair begins Friday and lasts through Sunday, September 14. The fair is the largest event in the state and attracts some 350,000 people. Each year since being elected to Congress, I have attended the fair to gather input and suggestions from Kansans to take back to Washington, D.C. The fair gives me an opportunity to meet many people from all parts of the state to discuss current issues. After these discussions, I always have a fresh perspective on the topics most important to Kansans.

As in years past, I will have a booth in the Pride of Kansas Building. Members of my staff will be available to answer questions and provide information. I will also take part in several events at the fair, including the annual Kansas Farm Bureau Leadership Breakfast on Saturday, September 6. I hope to see you and your family at the fair in Hutchinson.

Attending Labor Day Events

To celebrate Labor Day this year, I attended the 112th Annual Labor Day Celebration in Hoisington. Thanks to Brian Harrison for participating with me in this year's parade.

I also attended the City of Chapman Labor Day Parade. It was encouraging to see the city's recovery efforts and their commitment to continuing this parade which was first celebrated in 1909. Thanks to Phil Weishaar for organizing my participation in the parade.

In the Office

Alan and Phyllis Shurts of Beloit, Sydney and Mary Holmes of Hutchinson and Zack and Gavin Lake of Emporia came by the office this week for a tour of the U.S. Capitol.

Very truly yours,

Jerry

Director's Message

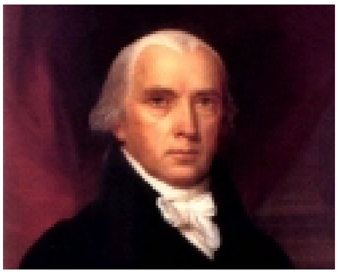
As the new year approaches, I'd like to reemphasize our commitment to presenting our statistics in plain language and easy-to-read formats as well as in more technical presentations. We've also tried to make it easier for our customers to access and download our statistics interactively. The Survey of Current Business, both the traditional printed publication and the online version, has been a big part of our efforts.

Over the past 12 months, we've made some improvements to the Survey. We've taken steps toward integrating our interactive data features with our traditional data presentations. We've enhanced our online Survey search engine. We've made our articles more Web friendly. And we've continued to publish *BEA Briefings* and *Research Spotlights* that explain important services and research in a less technical fashion.

We look forward to the new year. There are many initiatives underway that will result in new products and services from the Bureau of Economic Analysis. As always, the Survey will be a prime tool in communicating these developments to our users.

We appreciate your comments and questions.

J. Steven Landefeld
Director, Bureau of Economic Analysis



"It will be of little avail to the people that the laws are made by men of their own choice if the laws be so voluminous that they cannot be read, or so incoherent that they cannot be understood."

James Madison was the primary author of the Constitution and the fourth President of the United States. He was also a contributor to the Federalist Papers and a member of Congress from Virginia.

Summary of Major Points: Christopher B. Field, PhD[1], Carnegie Institution for Science[2], Energy and Commerce Committee, Subcommittee on Energy and Power: "Climate Science and the EPA's Greenhouse Gas Regulations", March 8, 2011

The modern understanding of climate change is based on many lines of robust, independent evidence, providing a foundation of well established conclusions, including the following, from the US Global Change Research Program: (*1) Global warming is unequivocal and primarily human-induced, (2) Climate changes are underway in the United States and are projected to grow, and (3) Widespread climate-related impacts are occurring now and are expected to increase.*

Against these foundations, I want to talk about the observed (not simulated) climate sensitivity of two important processes – US agriculture and wildfires in the Western US.

- Observed yields of corn, soybean, and cotton all have clear temperature thresholds, below which yields are stable and above which yields fall quickly with rising temperatures. A single day of 104˚F instead of 84˚F reduces corn yields by about 7%. Modest warming over the century is expected to reduce yields by 30-46% below levels that would otherwise occur, and severe warming could reduce yields by 63-82%
- Large wildfires in the Western US are more frequent, last longer, and occur over a longer fire season in years that are unusually warm. Based on observed patterns, a 1.8˚F warming would increase the annual area burned from 1.3 million acres (the average for 1970-2003) to 4.5 million acres.

[1] Any opinions, findings, conclusions, or recommendations expressed in this publication are those of the author and do not necessarily reflect those of the Carnegie Institution for Science or the IPCC

[2] The Carnegie Institution for Science is a not-for-profit organization dedicated to basic research for the benefit of humanity.

Statement of

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Before the

U.S. House of Representatives

Energy and Commerce Committee

Subcommittee on Energy and Power

"Climate Science and the EPA's Greenhouse Gas Regulations"

10:00 a.m., March 8, 2011

Room 2123, Rayburn House Office Building

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Climate Science and the EPA's Greenhouse Gas Regulations

Introduction

I thank Chairman Upton, Ranking Member Waxman, Chairman of the Subcommittee Whitfield, Ranking Member of the Subcommittee Rush, and the other Members of the Committee for the opportunity to speak with you today on observed impacts of climate on important processes in our country. My name is Christopher Field. I am director of the Department of Global Ecology at the Carnegie Institution for Science, a not-for-profit organization dedicated to basic research for the benefit of humanity. In addition, I am a professor in the Department of Environmental Earth System Science and the Department of Biology at Stanford University. Since September of 2008, I have served as co-chair of Working Group 2 of the Intergovernmental Panel on Climate Change. Working Group 2 is tasked with assessing scientific information concerning impacts of climate change, options for adaptation to climate changes that cannot be avoided, and vulnerability to climate change.

My personal research focuses on interactions among climate, the carbon cycle, and ecosystem processes, using approaches that range from ecosystem-scale climate manipulations to global climate models. I have published over 200 peer-reviewed papers in leading scientific journals, and was a coordinating lead author on the topic "North America" for the Working Group 2 contribution to the IPCC Fourth Assessment Report. I have served on many committees of the National Research Council and International Scientific Organizations. I am an elected member of the US National Academy of Sciences and the American Academy of Arts and Sciences as well as an elected Fellow of the American Association for the Advancement of Science.

In today's testimony, I will focus on two aspects of observed sensitivity of important processes to climate. The two processes are agricultural yields in the United States and wildfire in the Western United States. The sensitivities I want to discuss today are based on observations and not on simulations. All of the material I will be discussing today is based on publications in peer-reviewed scientific journals or on national or international assessments of thousands of scientific sources.

Robust Foundations of Current Knowledge

The starting point for the material I want to discuss today is a series of robust conclusions from climate science, synthesized in a number of recent major assessments including two 2010 reports from the US National Academy of Sciences, "Climate Stabilization Targets: Emissions, Concentrations, and Impacts over Decades to Millennia" (Solomon 2010), and "Advancing the Science of Climate Change" (Matson 2010), the 2009 report from the US Global Change Research Program, "Global Climate Change Impacts in the United States" (Karl et al. 2009) and the Fourth Assessment Report of the IPCC (IPCC 2007a, c, b). These documents provide a scientifically rich picture of a changing climate, the mechanisms that underlie observed and projected changes, impacts of climate change on individuals, ecosystems, economies, and regions, and the costs and benefits of changing practices to decrease the amount of climate change from a business-as-usual scenario.

The following 10 points, quoted from (Karl et al. 2009), form the foundation for any discussion of climate change and its impacts:

"*1. Global warming is unequivocal and primarily human-induced.*

Global temperature has increased over the past 50 years. This observed increase is due primarily to human induced emissions of heat-trapping gases.

2. Climate changes are underway in the United States and are projected to grow.

Climate-related changes are already observed in the United States and its coastal waters. These include increases in heavy downpours, rising temperature and sea level, rapidly retreating glaciers, thawing permafrost, lengthening growing seasons, lengthening ice-free seasons in the ocean and on lakes and rivers, earlier snowmelt, and alterations in river flows. These changes are projected to grow.

3. Widespread climate-related impacts are occurring now and are expected to increase.

Climate changes are already affecting water, energy, transportation, agriculture, ecosystems, and health. These impacts are different from region to region and will grow under projected climate change.

4. Climate change will stress water resources.

Water is an issue in every region, but the nature of the potential impacts varies. Drought, related to reduced precipitation, increased evaporation, and increased water loss from plants, is an important issue in many regions, especially in the West. Floods and water quality problems are likely to be amplified by climate change in most regions. Declines in mountain snowpack are important in the West and Alaska where snowpack provides vital natural water storage.

5. Crop and livestock production will be increasingly challenged.

Many crops show positive responses to elevated carbon dioxide and low levels of warming, but higher levels of warming often negatively affect growth and yields. Increased pests, water stress, diseases, and weather extremes will pose adaptation challenges for crop and livestock production.

6. Coastal areas are at increasing risk from sea-level rise and storm surge.

Sea-level rise and storm surge place many U.S. coastal areas at increasing risk of erosion and flooding, especially along the Atlantic and Gulf Coasts, Pacific Islands, and parts of Alaska. Energy and transportation infrastructure and other property in coastal areas are very likely to be adversely affected.

7. Risks to human health will increase.

Harmful health impacts of climate change are related to increasing heat stress, waterborne diseases, poor air quality, extreme weather events, and diseases transmitted by insects and rodents. Reduced cold stress provides some benefits. Robust public health infrastructure can reduce the potential for negative impacts.

8. Climate change will interact with many social and environmental stresses.

Climate change will combine with pollution, population growth, overuse of resources, urbanization, and other social, economic, and environmental stresses to create larger impacts than from any of these factors alone.

9. Thresholds will be crossed, leading to large changes in climate and ecosystems.
There are a variety of thresholds in the climate system and ecosystems. These thresholds determine, for example, the presence of sea ice and permafrost, and the survival of species, from fish to insect pests, with implications for society. With further climate change, the crossing of additional thresholds is expected.

10. Future climate change and its impacts depend on choices made today.
The amount and rate of future climate change depend primarily on current and future human-caused emissions of heat-trapping gases and airborne particles. Responses involve reducing emissions to limit future warming, and adapting to the changes that are unavoidable."

Recent Results: Observed Responses of the Temperature Sensitivity of US Agriculture

Globally and in the US, advancements in agriculture are among the crowning accomplishments of human ingenuity. Especially over the last century, yields have increased dramatically (Lobell et al. 2009), more than keeping pace with the growth of human population. One recent analysis concludes that agricultural intensification since 1961 has increased yields so much that the area in crops has not needed to change, even as demand has soared (Burney et al. 2010). As a consequence, intensification of agriculture has prevented deforestation that otherwise would have emitted 161 billion tons of carbon to the atmosphere.

Over recent decades, yields of most major crops have increased at 1-2% per year (Lobell and Field 2007), but an increasing body of evidence indicates that obtaining these yield increases is becoming more and more difficult, as climate change acts to resist or reverse yield increases from improvements in management and breeding. Using global records of yield trends in the world's six major food crops since 1961, my colleague David Lobell and I (Lobell and Field 2007) concluded that, at the global scale, effects of warming are already visible, with global yields of wheat, corn, and barley reduced since 1981 by 40 million tons per year below the levels that would occur without the warming. As of 2002 (the last year analyzed in the study), this represents an economic loss of approximately $5 billion per year.

In the United States, the observed temperature sensitivity of three major crops is even more striking. Based on a careful county-by county analysis of patterns of climate and yields of corn, soybeans, and cotton, Schlenker and Roberts (Schlenker and Roberts 2009) concluded that observed yields from all farms and farmers are relatively insensitive to temperature up to a threshold but fall rapidly as temperatures rise above the threshold. For farms in the United States, the temperature threshold is 84˚F for corn, 86˚F for soybeans, and 90˚F for cotton. For corn, a single day at 104˚F instead of 84˚F reduces observed yields by about 7%. These temperature sensitivities are based on observed responses, including data from all of the US counties that grow cotton and all of the Eastern counties that grow corn or soybeans. These are not simulated responses. They are observed in the aggregate yields of thousands of farms in thousands of locations.

The temperature sensitivity observed by Schlenker and Roberts (Schlenker and Roberts 2009) suggests a challenging future for US agriculture. Unless we can develop varieties with improved heat tolerance, modest warming (based on the IPCC B1 scenario) by the end of the 21st century will reduce yields by 30-46%. With a high estimate of climate change (based on the IPCC A1FI scenario), the loss of yield is 63-82%. These three major crops, in some ways the core of US agriculture, are exquisitely sensitive to warming. This result is very clear. We may be able to breed warming tolerant varieties, and it is possible that some of the yield losses due to warming will be compensated by positive responses to elevated atmospheric CO_2 (Long et al. 2006), but we will be trying to improve yields in a setting where warming is like an anchor pulling us back.

Recent Results: Observed Responses of the Temperature Sensitivity of Wildfires in the Western US

Wildfire is a common threat in the Western US. While historical, low-intensity wildfires can play an important role in sustaining the health of ecosystems (Minnich 1983), large, high-intensity wildfires destroy lives and homes, impact air quality, degrade watersheds, reduce economic activity, and eliminate wildlife habitat. In recent years, suppression costs have been over $1 billion per annum (Littell et al. 2009).

Westerling and colleagues (Westerling et al. 2006) compiled a database of 1166 large wildfires in the US during the period from 1970 to 2003. They observed a large increase in fire frequency, duration, and in the length of the fire season after the mid 1980s. This increase in fire activity was strongly related to warm spring and summer temperatures. Over this period, the length of time from snow-melt in the spring to the first snow-fall in the autumn is a good predictor of the fire risk.

Littell and colleagues (Littell et al. 2009) looked at a longer series of fire records and analyzed the data for relationships with temperature and rainfall. They found that, for most Western ecosystems, climate is a strong predictor of wildfire area burned, with current-year temperature explaining a significant amount of the variability in most regions. The striking feature of these results, based on observations and not simulations is the sensitivity. For most of the West, the annual area burned increases by 200 to over 600% for every 1.8˚F of warming (Solomon 2010). It is only in the driest regions, where warming makes it too dry for fire, that there is not a strong positive response. Based on these observed sensitivities, warming of 1.8˚F is expected to increase the area burned from 1.3 million acres per year (the 1970-2003 average) to 4.5 million acres (Solomon 2010).

Conclusion

Many lines of robust, independent evidence support the conclusion that the climate has been warming over the past century and that human emissions of heat-trapping gases are very likely responsible for much of the warming since the middle of the last century (IPCC 2007d). We are already seeing a wide range of impacts of these climate changes (IPCC 2007e). Although there is scientific uncertainty about the amount of future warming, it is very clear that, unless emissions of heat-trapping gases decrease dramatically, the planet will warm substantially in coming decades. New data on sensitivities, here I have discussed US agriculture and wildfire, emphasize the magnitude of the risk for the United States. Based on observations, not simulations, the sensitivities to warming of both agriculture and wildfire are more than sufficient to cause pervasive regional harm, with even modest warming.

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